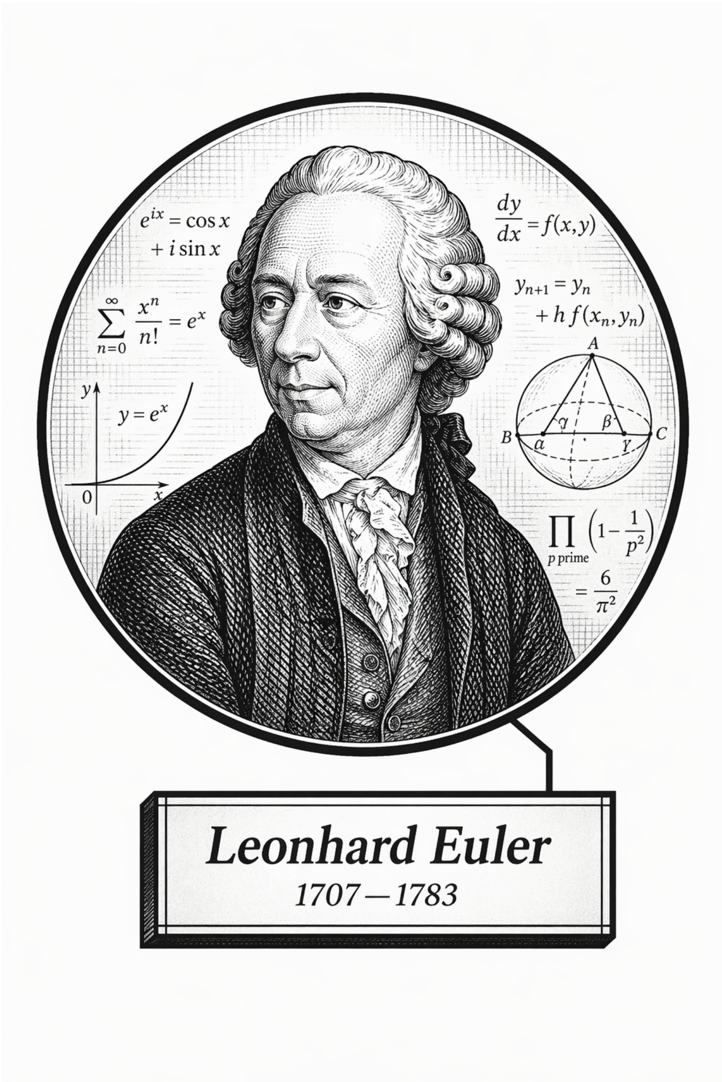
Contents

Volume VI

Computational Mathematics



Frontispiece placeholder.
Leonhard Euler, 1707–1783

Volume VI

Computational Mathematics

Self-Study Notes

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Part I

Computational Mathematics

Chapter 1

Calculus

calculus

Front Matter $\rightarrow$ **Calculus** $\rightarrow$ Geometric Modeling

1.1 Limits

Limits — Quick Reference

Limit of a function	$\lim_{x\to a} f(x) = L$: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < \lvert x - a\rvert < \delta \implies \lvert f(x) - L\rvert < \varepsilon$.
One-sided limits	$\lim_{x\to a^+} f(x)$: approach from the right; $\lim_{x\to a^-} f(x)$: approach from the left.
Limit exists	$\lim_{x\to a} f(x) = L$ iff $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = L$.
Key point	$f(a)$ need not be defined; the limit concerns behaviour *near* a, not *at* a.

1.2 Approximating Limits via Difference Quotients

Definition (Difference Quotient)

The *difference quotient* of a function f at $x = a$ with increment $h \neq 0$ is

$$\frac{f(a+h) - f(a)}{h}.$$

Remark (Geometric meaning). The difference quotient is the slope of the secant line through the points $(a,\ f(a))$ and $(a+h,\ f(a+h))$ on the graph of f. As $h \to 0$, the secant line rotates toward the tangent line at $(a, f(a))$, and the difference quotient approaches the slope of that tangent — provided the limit exists.

Remark (Approximating a limit numerically). For a function whose limit at a is difficult to evaluate algebraically, the difference quotient with small values of h provides a numerical estimate. Evaluating at $h = 0.1, 0.01, 0.001$ and observing the output stabilise gives evidence for the limiting value. For example, to estimate $\lim_{x\to 0} \dfrac{\sin x}{x}$:

h	$\sin(h)/h$
0.1	0.99833
0.01	0.99998
0.001	$0.99999\overline{9}$

The values converge to 1, consistent with the exact result $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$.

Remark (Limitation of numerical approximation). Numerical evidence is not a proof. Taking h small can introduce floating-point cancellation error, and a sequence of outputs that appears to stabilise may still diverge or converge to the wrong value for poorly behaved functions. The difference quotient is a computational tool for building intuition and checking algebra; the limit itself requires the ε-δ definition for a rigorous conclusion.

1.3 Limits of Functions

The Definition

Definition (Limit of a Function)

Let f be defined near a, except possibly at a itself. The *limit of $f(x)$ as x approaches a* is L, written

$$\lim_{x\to a} f(x) = L,$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Remark (Intuition). $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a from either side, without x reaching a. The ε names the required closeness of the output; the δ responds with the required closeness of the input.

Remark. The value $f(a)$ plays no role. The limit of $f(x) = \dfrac{x^2-1}{x-1}$ as $x \to 1$ is 2, even though the function is undefined at $x = 1$, because $\dfrac{x^2-1}{x-1} = x + 1$ for all $x \neq 1$.

Indeterminate Forms and Algebraic Techniques

Remark (Algebraic techniques). Direct substitution fails when it produces $\frac{0}{0}$ or $\frac{\infty}{\infty}$. The standard remedies are factoring, rationalising, and cancellation.

Example: Factoring

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x\to 1}(x+1) = 2.$$

Example: Rationalising

$$\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x} = \lim_{x\to 0} \frac{(\sqrt{x+1}-1)(\sqrt{x+1}+1)}{x(\sqrt{x+1}+1)} = \lim_{x\to 0} \frac{x}{x(\sqrt{x+1}+1)} = \frac{1}{2}.$$

The Squeeze Theorem

Theorem (Squeeze Theorem)

Let f, g, and h be functions on an open interval I containing c such that for all x in I,

$$f(x) \leq g(x) \leq h(x).$$

If

$$\lim_{x \to c} f(x) = L = \lim_{x \to c} h(x),$$

then

$$\lim_{x \to c} g(x) = L.$$

Remark (Intuition). g is trapped between f and h. If both outer functions converge to the same value L, the middle function has nowhere else to go.

Remark (Canonical application). The standard use is $\lim_{x \to 0} \dfrac{\sin x}{x} = 1$, established by bounding $\sin x/x$ between $\cos x$ and 1 and applying the theorem as $x \to 0$.

Limits of Functions Equal at All But One Point

Theorem (Limits of Functions Equal at All But One Point)

Let $g(x) = f(x)$ for all x in an open interval containing c, except possibly at c itself, and let $\lim_{x \to c} g(x) = L$. Then

$$\lim_{x \to c} f(x) = L.$$

Remark. This theorem licenses the standard algebraic technique of cancelling a common factor that vanishes at c — such as the $(x-1)$ factor in $\dfrac{x^2-1}{x-1}$ — and then evaluating the simplified expression. The original function and the simplified function agree everywhere except at c, so they share the same limit there.

One-Sided Limits

Definition (One-Sided Limits)

Left-hand limit. Let f be defined on (a, c) for some $a < c$. The *left-hand limit* of f at c is L, written

$$\lim_{x \to c^-} f(x) = L,$$

if for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x \in (a, c)$, if $|x - c| < \delta$ then $|f(x) - L| < \varepsilon$.

Right-hand limit. Let f be defined on (c, b) for some $b > c$. The *right-hand limit* of f at c is L, written

$$\lim_{x \to c^+} f(x) = L,$$

if for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x \in (c, b)$, if $|x - c| < \delta$ then $|f(x) - L| < \varepsilon$.

Remark. The only difference from the two-sided definition is the restriction of x to one side of c. One-sided limits are the natural tool for piecewise-defined functions, where the formula changes at c.

Limits and One-Sided Limits

Theorem (Limits and One-Sided Limits)

Let f be defined on an open interval I containing c, except possibly at c. Then

$$\lim_{x\to c} f(x) = L$$

if, and only if,

$$\lim_{x\to c^-} f(x) = L \quad \text{and} \quad \lim_{x\to c^+} f(x) = L.$$

Remark. This theorem is the standard test for non-existence of a limit: if the left- and right-hand limits exist but differ, the two-sided limit does not exist. The canonical example is $f(x) = |x|/x$ at $x = 0$, where $\lim_{x\to 0^-} = -1$ and $\lim_{x\to 0^+} = 1$.

1.4 Limit Properties

Theorem (Basic Limit Properties)

Let b, c, L, and K be real numbers, let n be a positive integer, and let f and g be functions defined on an open interval I containing c with

$$\lim_{x\to c} f(x) = L \qquad \lim_{x\to c} g(x) = K.$$

The following limits hold.

Constants	$\lim_{x\to c} b = b$
Identity	$\lim_{x\to c} x = c$
Sums/Differences	$\lim_{x\to c}(f(x) \pm g(x)) = L \pm K$
Scalar Multiples	$\lim_{x\to c}(b \cdot f(x)) = bL$
Products	$\lim_{x\to c}(f(x) \cdot g(x)) = LK$
Quotients	$\lim_{x\to c}(f(x)/g(x)) = L/K, \quad K \neq 0$
Powers	$\lim_{x\to c} f(x)^n = L^n$
Roots	$\lim_{x\to c} \sqrt[n]{f(x)} = \sqrt[n]{L}$ (If n is even, require $f(x) \geq 0$ on I.)
Compositions	If additionally $\lim_{x\to L} g(x) = K$ and $g(L) = K$, then $\lim_{x\to c} g(f(x)) = K$.

Remark. For polynomials and rational functions (where the denominator is nonzero), the limit laws reduce to direct substitution: $\lim_{x \to a} p(x) = p(a)$.

Remark. The proofs of these properties follow from the ε-δ definition and are developed in the analysis chapters. Used together, they reduce the limit of any rational function at a point where the denominator is nonzero to direct substitution.

1.5 Limits Involving Infinity

Infinite Limits

Definition (Limits of Infinity)

Let I be an open interval containing c, and let f be defined on I, except possibly at c.

- The limit of $f(x)$ as x approaches c is *infinity*, denoted

$$\lim_{x \to c} f(x) = \infty,$$

  if given any $N > 0$, there exists $\delta > 0$ such that for all $x \in I$ with $x \neq c$, if $|x - c| < \delta$ then $f(x) > N$.

- The limit of $f(x)$ as x approaches c is *negative infinity*, denoted

$$\lim_{x \to c} f(x) = -\infty,$$

  if given any $N < 0$, there exists $\delta > 0$ such that for all $x \in I$ with $x \neq c$, if $|x - c| < \delta$ then $f(x) < N$.

Remark (Intuition). N plays the role that ε plays in the standard definition: the adversary names an arbitrarily large threshold, and the response is a δ-neighbourhood of c on which the function exceeds it. Infinity is not a value being approached — it is a description of unbounded growth.

One-Sided Limits of Infinity

Definition (One-Sided Limits of Infinity)

Left-hand limit. Let f be defined on (a, c) for some $a < c$. The left-hand limit of f at c is infinity,

$$\lim_{x \to c^-} f(x) = \infty,$$

if given any $N > 0$, there exists $\delta > 0$ such that for all $a < x < c$, if $|x - c| < \delta$ then $f(x) > N$.

Right-hand limit. Let f be defined on (c, b) for some $b > c$. The right-hand limit of f at c is infinity,

$$\lim_{x \to c^+} f(x) = \infty,$$

if given any $N > 0$, there exists $\delta > 0$ such that for all $c < x < b$, if $|x - c| < \delta$ then $f(x) > N$.

The left- and right-hand limits of $-\infty$ are defined analogously, replacing $f(x) > N$ with $f(x) < N$ for $N < 0$.

Vertical Asymptotes

Definition (Vertical Asymptote)

Let I be an interval that either contains c or has c as an endpoint, and let f be defined on I, except possibly at c.
If the limit of $f(x)$ as x approaches c from either the left or the right (or both) is ∞ or $-\infty$, then the line $x = c$ is a *vertical asymptote* of f.

Remark. The standard example is $f(x) = 1/x$, which has a vertical asymptote at $x = 0$: $\lim_{x\to 0^-} 1/x = -\infty$ and $\lim_{x\to 0^+} 1/x = +\infty$. The two one-sided limits need not agree — or even have the same sign — for $x = c$ to qualify as a vertical asymptote.

Indeterminate Forms

Definition (Indeterminate Forms)

An *indeterminate form* arises when substituting the limit value into an expression produces a form whose limiting value cannot be determined from the form alone. The standard indeterminate forms are:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 0^0, \quad 1^\infty, \quad \infty^0.$$

Remark. An indeterminate form is not an error — it is a signal that more work is required. The form $0/0$ is the most common: it appears whenever a common factor vanishes at the limit point (handled by cancellation) or in derivatives via the difference quotient. The forms $0/0$ and ∞/∞ are handled by L'Hôpital's Rule; the others are typically reduced to one of these two by algebraic manipulation.

Remark (Common error). The forms $0/0$ and ∞/∞ are indeterminate, but $c/0$ for $c \neq 0$ is *not* indeterminate — it indicates an infinite limit (a vertical asymptote), not an unresolved expression. Similarly, $\infty + \infty = \infty$ is determinate; only $\infty - \infty$ is not.

Limits at Infinity and Horizontal Asymptotes

Definition (Limits at Infinity and Horizontal Asymptotes)

Let L be a real number.

1. Let f be defined on (a, ∞) for some a. The *limit of f at infinity* is L, denoted $\lim_{x\to\infty} f(x) = L$, if for every $\varepsilon > 0$ there exists $M > a$ such that if $x > M$ then $|f(x) - L| < \varepsilon$.
2. Let f be defined on $(-\infty, b)$ for some b. The *limit of f at negative infinity* is L, denoted $\lim_{x\to-\infty} f(x) = L$, if for every $\varepsilon > 0$ there exists $M < b$ such that if $x < M$ then $|f(x) - L| < \varepsilon$.
3. If $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$, the line $y = L$ is a *horizontal asymptote* of f.

Remark. A function can have at most two horizontal asymptotes (one as $x \to +\infty$ and one as $x \to -\infty$), and they need not be equal. For example, $f(x) = \arctan(x)$ has horizontal asymptotes at $y = \pi/2$ and $y = -\pi/2$.

Limits of Rational Functions at Infinity

Theorem (Limits of Rational Functions at Infinity)

Let $f(x)$ be a rational function of the form

$$f(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0},$$

where m, n are positive integers and $a_n, b_m \neq 0$. Then:

1. If $n = m$, then $\displaystyle\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = \frac{a_n}{b_m}$.
2. If $n < m$, then $\displaystyle\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = 0$.
3. If $n > m$, then $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$ are both infinite.

Remark. The rule follows from dividing numerator and denominator by x^m and observing that all terms with negative powers of x vanish as $x \to \pm\infty$, leaving only the ratio $a_n x^{n-m}/b_m$. When $n = m$ this ratio is the constant a_n/b_m; when $n < m$ it tends to zero; when $n > m$ it grows without bound.

1.6 Continuity

1.7 Continuity

Definition

Definition (Continuous Function)

Let f be a function whose domain contains an open interval I.

1. f is *continuous at a point* $c \in I$ if $\lim_{x \to c} f(x) = f(c)$.
2. f is *continuous on* I if f is continuous at every $c \in I$. If f is continuous on $(-\infty, \infty)$, f is said to be *continuous everywhere.*

Remark (Three-condition test). f is continuous at c if and only if all three of the following hold:

1. $\lim_{x \to c} f(x)$ exists,
2. $f(c)$ is defined, and
3. $\lim_{x \to c} f(x) = f(c)$.

The three conditions make explicit what can go wrong: the limit may fail to exist, the function may be undefined at c, or the two values may disagree.

Continuity on Closed Intervals

Definition (Continuity on Closed Intervals)

Let f be defined on the closed interval $[a, b]$. f is *continuous on* $[a, b]$ if:

1. f is continuous on (a, b),
2. $\lim_{x \to a^+} f(x) = f(a)$, and
3. $\lim_{x \to b^-} f(x) = f(b)$.

Remark. Condition (2) is called *continuity from the right* at a; condition (3) is *continuity from the left* at b. The same idea applies to half-open intervals $[a, b)$ and $(a, b]$.

Classification of Discontinuities

Definition (Types of Discontinuity)

When f fails to be continuous at a, the discontinuity falls into one of three categories.

1. **Removable discontinuity.** $\lim_{x\to a} f(x) = L$ exists, but either $L \neq f(a)$ or $f(a)$ is undefined. The discontinuity is "removable" because redefining $f(a) = L$ produces a function continuous at a that agrees with f everywhere else.

2. **Jump discontinuity.** $\lim_{x\to a^+} f(x) = L$ and $\lim_{x\to a^-} f(x) = M$ both exist, but $L \neq M$. The two-sided limit does not exist; the graph visually "jumps" from one value to another as x crosses a.

3. **Infinite discontinuity.** f is unbounded near $x = a$: the value of f becomes arbitrarily large (or large and negative) as x approaches a. At least one one-sided limit is $\pm\infty$, and $x = a$ is a vertical asymptote of f.

Remark (Canonical examples). Removable: $f(x) = \dfrac{x^2-1}{x-1}$ at $x = 1$ — the limit is 2 but $f(1)$ is undefined. Jump: $f(x) = \lfloor x \rfloor$ at every integer — the left- and right-hand limits differ by 1. Infinite: $f(x) = \dfrac{1}{x}$ at $x = 0$ — the function grows without bound from both sides.

Properties of Continuous Functions

Theorem (Properties of Continuous Functions)

Let f and g be continuous on an interval I, let $c \in \mathbb{R}$, and let n be a positive integer. The following functions are continuous on I:

Sums/Differences	$f \pm g$
Constant Multiple	$c \cdot f$
Products	$f \cdot g$
Quotients	f/g (provided $g \neq 0$ on I)
Powers	f^n
Roots	$\sqrt[n]{f}$ (if n is even, require $f \geq 0$ on I)
Compositions	If f is continuous on I with range J, and g is continuous on J, then $g \circ f$ is continuous on I.

Remark. Polynomials are continuous everywhere. Rational functions are continuous on any interval that avoids the zeros of the denominator.

Continuous Functions

Theorem (Continuous Functions)

The following functions are continuous on their domains:

$\sin(x)$	$\cos(x)$
$\tan(x)$	$\cot(x)$
$\sec(x)$	$\csc(x)$
$\ln(x)$	$a^x \quad (a > 0)$
$\sqrt[n]{x}$	(n a positive integer)

The Intermediate Value Theorem

Theorem (Intermediate Value Theorem)

Let f be continuous on $[a, b]$ and, without loss of generality, let $f(a) < f(b)$. Then for every value y with $f(a) < y < f(b)$, there exists at least one $c \in (a, b)$ such that $f(c) = y$.

Remark (Intuition). A continuous function cannot skip values. If f is at height $f(a)$ on the left and height $f(b)$ on the right, it must pass through every height in between at least once.

Remark (Root-finding application). The IVT is the theoretical foundation of the *Bisection Method*. If $f(a) < 0$ and $f(b) > 0$ with f continuous on $[a, b]$, then there is at least one root $c \in (a, b)$ where $f(c) = 0$. Evaluating f at the midpoint $d = (a + b)/2$ and replacing whichever endpoint has the same sign as $f(d)$ halves the interval at each step, converging to the root.

Remark (What the IVT does not say). The theorem guarantees existence of at least one c, but gives no information about how many such values exist or where they are. It also requires continuity on the *entire* closed interval $[a, b]$; a single discontinuity is enough to invalidate the conclusion.

Chapter 2

Geometric Modeling

geometric-modeling

Calculus $\rightarrow$ **Geometric Modeling** $\rightarrow$ Computational Geometry

2.1 Curves and Surfaces

Parametric Curves and Surfaces — Quick Reference

Curve	$C(u) : [a, b] \rightarrow \mathbb{R}^n$; vector-valued, one parameter.
Velocity	$C'(u)$; tangent direction and speed of traversal.
Unit tangent	$\hat{T}(u) = C'(u)/\|C'(u)\|$ at a regular point.
Surface	$S(u, v) : R \rightarrow \mathbb{R}^3$; vector-valued, two parameters.
Partial derivatives	$S_u = \partial S/\partial u$, $S_v = \partial S/\partial v$; tangent to isoparametric lines.
Unit normal	$N = (S_u \times S_v)/\|S_u \times S_v\|$ at a regular point.
Tangent plane	Spanned by S_u and S_v at the base point.
Directional deriv.	$D_{\mathbf{v}}S = \nabla S \cdot \hat{\mathbf{v}}$; rate of change in direction $\mathbf{v}$.

2.2 Parametric Curves and Surfaces

Parametric Curves

Definition (Parametric Curve)

A *parametric curve* in $\mathbb{R}^n$ is a vector-valued function

$$C : [a, b] \rightarrow \mathbb{R}^n, \qquad C(u) = \big(x_1(u), x_2(u), \ldots, x_n(u)\big),$$

where each coordinate function $x_i(u)$ is a real-valued function of the scalar parameter u. The interval $[a, b]$ is typically normalised to $[0, 1]$.

Definition (Velocity, Regularity, Unit Tangent)

The *velocity vector* (or *tangent vector*) at u is the derivative

$$C'(u) = \big(x_1'(u),\, x_2'(u),\, \ldots,\, x_n'(u)\big).$$

The curve is *regular* at u if $C'(u) \neq \mathbf{0}$. At a regular point, the *unit tangent vector* is

$$\hat{T}(u) = \frac{C'(u)}{\|C'(u)\|}.$$

The scalar $\|C'(u)\|$ is the *speed* of traversal.

Remark (Properties of parametric curves). Parametric curves carry structure that implicit curves do not:

- **Direction.** The parameter increases from a to b, giving the curve a natural orientation. Reversing the parametrisation reverses the direction.
- **Velocity and speed.** $C'(u)$ measures both the direction of motion and how fast the curve is being traversed. A curve can be *reparametrised* (change of variable $u = \phi(t)$) without changing its shape; the velocity changes but the trace does not.
- **Endpoint interpolation.** The start and end points are $C(a)$ and $C(b)$, accessible directly.
- **Boundedness.** The curve is defined on a closed interval, so it represents a bounded segment, not an infinite line.
- **Cusps.** Where $C'(u) = \mathbf{0}$, the curve may have a cusp (velocity drops to zero, direction may reverse). These are degenerate points excluded by the regularity condition.

Parametric Surfaces

Definition (Parametric Surface)

A *parametric surface* in $\mathbb{R}^3$ is a vector-valued function of two parameters,

$$S : R \longrightarrow \mathbb{R}^3, \qquad S(u,v) = \big(x(u,v),\; y(u,v),\; z(u,v)\big),$$

where $R \subseteq \mathbb{R}^2$ is the *parameter domain* (typically the unit square $[0,1]^2$).

Partial Derivatives and Isoparametric Curves

Definition (Partial Derivative Vectors)

The *partial derivative vectors* of S at (u, v) are

$$S_u(u, v) = \frac{\partial S}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right), \qquad S_v(u, v) = \frac{\partial S}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right).$$

S_u is the tangent vector to the surface in the u-direction: it is the velocity of the curve obtained by fixing v and varying u. S_v is the corresponding tangent vector in the v-direction.

Definition (Isoparametric Curves)

Fixing one parameter on $S(u, v)$ produces a curve on the surface:

- *u-curve* at $v = v_0$: $C_{v_0}(u) = S(u,\, v_0)$, tangent $= S_u(u, v_0)$.
- *v-curve* at $u = u_0$: $C_{u_0}(v) = S(u_0,\, v)$, tangent $= S_v(u_0, v)$.

These *isoparametric curves* (isocurves) sweep the surface and form its parameter grid. The two families intersect at every surface point $S(u_0, v_0)$.

Directional Derivative

Definition (Directional Derivative on a Surface)

The *directional derivative* of S at (u_0, v_0) in the direction $\mathbf{d} = (d_u, d_v)$ (a unit vector in parameter space) is

$$D_{\mathbf{d}}\, S = d_u\, S_u(u_0, v_0) + d_v\, S_v(u_0, v_0).$$

It measures the rate of change of position on the surface as the parameter moves in direction $\mathbf{d}$ in the uv-plane. Every tangent vector to the surface at (u_0, v_0) is a directional derivative for some choice of $\mathbf{d}$.

Unit Normal and Tangent Plane

Definition (Regularity, Unit Normal, Tangent Plane)

The surface is *regular* at (u_0, v_0) if $S_u \times S_v \neq \mathbf{0}$ there.
At a regular point, the *unit normal vector* is

$$N(u_0, v_0) = \frac{S_u(u_0, v_0) \;\times\; S_v(u_0, v_0)}{\|S_u(u_0, v_0) \;\times\; S_v(u_0, v_0)\|}\,.$$

The *tangent plane* at $S(u_0, v_0)$ is the plane through $S(u_0, v_0)$ spanned by S_u and S_v:

$$\Pi \;=\; \big\{\, S(u_0, v_0) + \alpha\, S_u + \beta\, S_v \;\big|\; \alpha, \beta \in \mathbb{R} \,\big\}.$$

Its equation in terms of position $\mathbf{p}$ is

$$N \cdot \big(\mathbf{p} - S(u_0, v_0)\big) = 0.$$

Remark (Geometric meaning of $S_u \times S_v$). The magnitude $\|S_u \times S_v\|$ equals the area of the parallelogram spanned by S_u and S_v, which is the *local area element* of the surface. Where $\|S_u \times S_v\| = 0$ the surface is degenerate (the two tangent directions are parallel or one vanishes), and the normal is not defined.

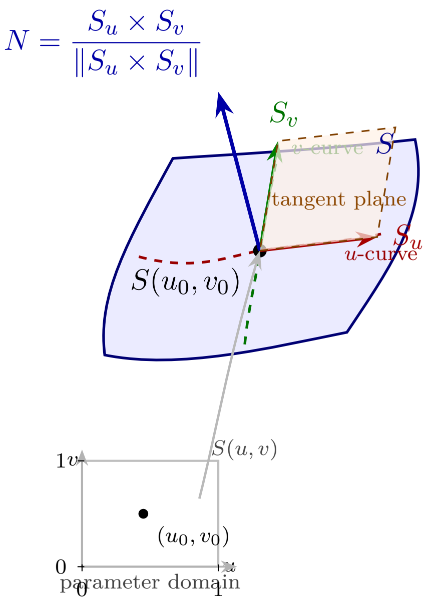

Figure: A parametric surface patch $S(u, v)$ with its parameter domain (lower left). At the point $S(u_0, v_0)$, the partial derivative vectors S_u (red) and S_v (green) are tangent to the isoparametric curves; together they span the tangent plane (orange). The unit normal $N = S_u \times S_v \,/\, \|S_u \times S_v\|$ (blue) is perpendicular to the tangent plane.

Status: Active

This section is actively expanding alongside work in Piegl & Tiller, *The NURBS Book*. Subsequent sections on Bézier curves, B-splines, NURBS, and continuity conditions follow.

2.3 Bezier Curves

Chapter 3

Computational Geometry

computational-geometry

Geometric Modeling $\rightarrow$ **Computational Geometry** $\rightarrow$ Computational Linear Algebra

3.1 Convex Hulls

Toolkit: Convex Hulls

Ambient space	Finite point sets in $\mathbb{R}^2$.
Convexity	Line segments and convex combinations.
Certificates	Extreme points, supporting lines, and maximizing directions.
Order data	Distinct x-coordinates and slopes between ordered points.
Algorithmic shape	Build the upper hull by repeatedly choosing the next maximal slope.

Remark (Exposition). Computational geometry studies finite geometric data, such as point sets in the plane, using exact definitions and algorithms. Convex hulls are the first organizing example: they connect geometry, order and maximization, supporting lines, and algorithmic construction.

3.2 Convex Hulls

Convexity and Hulls

Convexity and Convex Hulls

This formulation makes precise the geometric idea of filling in all straight segments forced by a point set. Convex combinations give an algebraic language for the same operation, and the convex hull records the smallest set closed under that operation.

Definition

Definition 3.2.1 (Convex Set). A set $S \subseteq \mathbb{R}^2$ is *convex* if for every pair of points $a, b \in S$ and every $\lambda \in [0, 1]$,

$$\lambda a + (1 - \lambda)b \in S.$$

Equivalently, S contains the line segment between any two of its points.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Definition

Definition 3.2.2 (Convex Combination). A *convex combination* of points $q_1, \ldots, q_n \in \mathbb{R}^2$ is a point of the form

$$\sum_{i=1}^{n} \lambda_i q_i, \qquad \lambda_i \geq 0 \text{ for all } i, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

The numbers λ_i are called the weights.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Definition

Definition 3.2.3 (Convex Hull). The *convex hull* of a set $P \subseteq \mathbb{R}^2$, written $\operatorname{conv}(P)$, is the set of all convex combinations of finitely many points of P:

$$\operatorname{conv}(P) = \left\{ \sum_{i=1}^{n} \lambda_i q_i \;\middle|\; n \geq 1,\ q_i \in P,\ \lambda_i \geq 0,\ \sum_{i=1}^{n} \lambda_i = 1 \right\}.$$

Equivalently, $\operatorname{conv}(P)$ is the smallest convex set containing P, i.e. the intersection of all convex sets that contain P.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Lemma

Lemma 3.2.4 (Convex hull as the smallest convex set). *Let* $P \subseteq \mathbb{R}^2$*. Then* $\operatorname{conv}(P)$ *is convex, contains* P*, and is contained in every convex set that contains* P*. Hence*

$$\operatorname{conv}(P) = \bigcap \{K \subseteq \mathbb{R}^2 : K \textit{ is convex and } P \subseteq K\}.$$

Remark (Proof). See the proof stub.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Remark (Dependencies).

- Convex Set
- Convex Combination
- Convex Hull

Supporting Functionals

Supporting Functionals and Vertices

This topic isolates the passage from geometry to optimization. A supporting line is a geometric certificate that a convex set lies on one side of a boundary, while a linear functional turns that certificate into a maximum.

Definition

Definition 3.2.5 (Extreme Point)**.** Let $S \subseteq \mathbb{R}^2$ be convex. A point $p \in S$ is an *extreme point* of S if whenever

$$p = \lambda a + (1-\lambda)b, \qquad a, b \in S, \quad \lambda \in (0,1),$$

it follows that $a = b = p$. For a finite point set P, the extreme points of $\operatorname{conv}(P)$ are called the vertices of the convex hull.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Definition

Definition 3.2.6 (Supporting Line)**.** A line $\ell \subseteq \mathbb{R}^2$ is a *supporting line* of a convex set S at a point $p \in S$ if $p \in \ell$ and S lies entirely in one of the two closed half-planes bounded by ℓ. Equivalently, there exists $d \in \mathbb{R}^2 \setminus \{0\}$ such that

$$\ell = \{x \in \mathbb{R}^2 : \langle d, x \rangle = \langle d, p \rangle\}$$

and

$$\langle d, q \rangle \leq \langle d, p \rangle \qquad \text{for all } q \in S.$$

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Lemma

Lemma 3.2.7 (Linear functionals preserve convex combinations). *Let $d \in \mathbb{R}^2$, and let*

$$z = \sum_{i=1}^{n} \lambda_i q_i$$

be a convex combination of points $q_1, \ldots, q_n \in \mathbb{R}^2$. Then

$$\langle d, z \rangle = \sum_{i=1}^{n} \lambda_i \langle d, q_i \rangle.$$

Consequently, if $\langle d, q_i \rangle \le M$ for every i, then

$$\langle d, z \rangle \le M.$$

Remark (Proof). See the proof stub.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Theorem

Remark (Dependencies).

- Convex Combination
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Theorem 3.2.8 (Supporting functional characterization of hull vertices). *Let $P \subset \mathbb{R}^2$ be finite and in general position. A point $p \in P$ is a vertex of* $\mathrm{conv}(P)$ *if and only if there exists a direction $d \in \mathbb{R}^2 \setminus \{0\}$ such that*

$$\langle d, p \rangle > \langle d, q \rangle \qquad \textit{for all } q \in P \setminus \{p\}.$$

Equivalently, p is the unique point of P attaining

$$\max_{q \in P} \langle d, q \rangle.$$

Thus every hull vertex is a maximum of P under some linear functional.

Remark (Proof). See the proof stub.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Remark (Finite maximum form). The maximum is used here because P is finite. This is the finite-set version of the same supporting-functional intuition that appears in supremum language for more general sets.

Remark (Dependencies).

- Convex Combination
- Convex Hull
- Extreme Point
- Supporting Line
- Linear Functionals Preserve Convex Combinations

Upper Hull Structure

Monotone Slopes and Greedy Construction

This topic converts the supporting-line description into an ordered planar construction. Under the general-position assumptions, each relevant maximum is unique, and the upper hull can be recognized by the strict monotonicity of successive slopes.

Assumption 3.2.9 (General Position for the Hull Theorems)**.** Let $P \subset \mathbb{R}^2$ be a finite set in general position: the points have distinct x-coordinates and no three points are collinear. For $p \in \mathbb{R}^2$, write $x(p)$ and $y(p)$ for its coordinates.

Theorem

Theorem 3.2.10 (Monotone slope characterization of the upper hull)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. Let*

$$v_0, v_1, \ldots, v_m$$

be the vertices of the upper hull of P, listed in increasing x-coordinate, and define

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \qquad k = 0, 1, \ldots, m-1.$$

Then

$$s_0 > s_1 > \cdots > s_{m-1}.$$

Conversely, if points $w_0, \ldots, w_m$ satisfy

$$x(w_0) < x(w_1) < \cdots < x(w_m)$$

and their consecutive slopes are strictly decreasing, then every w_k is a vertex of

$$\operatorname{conv}(\{w_0, \ldots, w_m\}).$$

The lower hull is characterized similarly, with strictly increasing consecutive slopes.

Remark (Proof). See the proof stub.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Theorem

Remark (Dependencies).

- Convex Hull
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- Supporting Functional Characterization of Hull Vertices

Theorem 3.2.11 (Greedy slope construction of the upper hull). *Let $P \subset \mathbb{R}^2$ be finite and in general position. Define*

$$v_0 = \arg\min_{p \in P} x(p).$$

Given v_k with

$$x(v_k) < \max_{p \in P} x(p),$$

define

$$v_{k+1} = \arg\max_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

Terminate once

$$v_k = \arg\max_{p \in P} x(p).$$

Then the points

$$v_0, v_1, \ldots, v_m$$

produced by this construction are exactly the vertices of the upper hull of P, and the slopes encountered along the way are strictly decreasing.

Remark (Proof). See the proof stub.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Remark (Uniqueness of choices). The distinct x-coordinates make the initial and terminal x-extreme points unique. The no-three-collinear condition makes the slope maximizer in the greedy step unique.

Remark (Proof roadmap). The proof ladder is:

convex combinations $\to$ linear functionals $\to$ supporting lines $\to$ monotone slopes $\to$ greedy hull const

The first two lemmas are algebraic. The monotone slope theorem is an analytic-geometry statement about three ordered points. The greedy construction theorem is the algorithmic consequence.

Remark (Dependencies).

- Convex Hull
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- Supporting Functional Characterization of Hull Vertices
- Monotone slope characterization of the upper hull

Chapter 4

Computational Linear Algebra

computational-linear-algebra

Computational Geometry $\rightarrow$ **Computational Linear Algebra** $\rightarrow$ Computational Probability

4.1 Vectors

Vectors and Linear Combinations — Quick Reference

Vector in $\mathbb{R}^n$	An ordered list $(v_1, v_2, \ldots, v_n)$ of n real numbers.
Addition	$(v_1, \ldots, v_n) + (w_1, \ldots, w_n) = (v_1 + w_1, \ldots, v_n + w_n)$.
Scalar multiplication	$c(v_1, \ldots, v_n) = (cv_1, \ldots, cv_n)$.
Linear combination	$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$; scalars times vectors, summed.
Convex combination	Linear combination with $c_i \geq 0$ and $\sum c_i = 1$; result lies inside the convex hull.
Standard basis vectors	$\mathbf{e}_j \in \mathbb{R}^n$: all zeros except a 1 in position j. Every vector is a linear combination of $\mathbf{e}_1, \ldots, \mathbf{e}_n$.

Matrices — Quick Reference

Matrix	Rectangular array of numbers; $A \in \mathbb{R}^{m \times n}$ has m rows, n columns.
Matrix–vector	$(A\mathbf{x})_i = \sum_j A_{ij}x_j$; $A\mathbf{x}$ is a linear combination of columns of A.
Matrix–matrix	$(AB)_{ij} = \sum_k A_{ik}B_{kj}$; requires inner dimensions to match.
Transpose	$(A^T)_{ij} = A_{ji}$; rows become columns.
Identity	I_n: ones on diagonal, zeros elsewhere; $IA = AI = A$.
Column view	$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$ where $\mathbf{a}_j$ are the columns of A.

Lengths and Dot Products — Quick Reference

Dot product	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n \in \mathbb{R}$.
Length (norm)	$\lVert \mathbf{v} \rVert = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}$.
Unit vector	$\hat{\mathbf{v}} = \mathbf{v}/\lVert \mathbf{v} \rVert$; has length 1.
Angle	$\cos\theta = \dfrac{\mathbf{v} \cdot \mathbf{w}}{\lVert \mathbf{v} \rVert \, \lVert \mathbf{w} \rVert}$.
Orthogonality	$\mathbf{v} \perp \mathbf{w}$ iff $\mathbf{v} \cdot \mathbf{w} = 0$.
Cauchy–Schwarz	$\lvert \mathbf{v} \cdot \mathbf{w} \rvert \le \lVert \mathbf{v} \rVert \, \lVert \mathbf{w} \rVert$.

4.2 Vectors and Linear Combinations

Vectors

Definition (Vector in $\mathbb{R}^n$)

A *vector* in $\mathbb{R}^n$ is an ordered list of n real numbers, written $\mathbf{v} = (v_1, v_2, \ldots, v_n)$. The number n is the *dimension* of the vector.

Vector addition:

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1,\ v_2 + w_2,\ \ldots,\ v_n + w_n).$$

Scalar multiplication:

$$c\mathbf{v} = (cv_1,\ cv_2,\ \ldots,\ cv_n), \quad c \in \mathbb{R}.$$

Remark (Geometric meaning). In $\mathbb{R}^2$ and $\mathbb{R}^3$, a vector is an arrow indicating displacement. Its entries specify length and direction, but *not* location — the same vector can be placed anywhere in space. Addition corresponds to following one displacement and then another (head-to-tail rule). Scalar multiplication stretches or shrinks the arrow; negative scalars reverse its direction.

Theorem (Properties of Vector Operations)

For all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$:

1. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity)
2. $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$ (associativity)
3. $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$ (distributivity over vector addition)
4. $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$ (distributivity over scalar addition)
5. $c(d\mathbf{v}) = (cd)\mathbf{v}$ (compatibility)

Linear Combinations

Definition (Linear Combination)

A *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ is any vector of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k,$$

where $c_1, c_2, \ldots, c_k \in \mathbb{R}$ are called the *coefficients*.
A *convex combination* is a linear combination in which $c_i \geq 0$ for all i and $\sum_{i=1}^{k} c_i = 1$.

Remark (Why convex combinations matter for NURBS). Every point on a Bézier or B-spline curve is a convex combination of control points. The Bernstein basis functions $B_{i,n}(u)$ are non-negative and sum to 1, so $C(u) = \sum_{i=0}^{n} B_{i,n}(u)P_i$ is a convex combination of the control points $\{P_i\}$ at every parameter value u. This is the *convex hull property*: the curve lies entirely within the convex hull of its control polygon.

Definition (Standard Basis Vectors)

For $j = 1, 2, \ldots, n$, the *j-th standard basis vector* $\mathbf{e}_j \in \mathbb{R}^n$ has a 1 in position j and 0s everywhere else. Every vector $\mathbf{v} = (v_1, \ldots, v_n) \in \mathbb{R}^n$ can be written as

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n.$$

4.3 Lengths and Dot Products

The Dot Product

Definition (Dot Product)

The *dot product* (or *inner product*) of $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is

$$\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2 + \cdots + v_nw_n \ \in \ \mathbb{R}.$$

Theorem (Properties of the Dot Product)

For all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

1. $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$ (commutativity)
2. $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$ (distributivity)
3. $\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$ (scalar compatibility)
4. $\mathbf{v} \cdot \mathbf{v} \geq 0$, with equality iff $\mathbf{v} = \mathbf{0}$ (positive definiteness)

Length and Angle

Definition (Length and Unit Vector)

The *length* (or *norm*) of $\mathbf{v} \in \mathbb{R}^n$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

A *unit vector* has length 1. The *normalisation* of a nonzero vector $\mathbf{v}$ is $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$, which is the unit vector in the direction of $\mathbf{v}$.

Theorem (Cauchy–Schwarz Inequality)

For all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \, \|\mathbf{w}\|,$$

with equality if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel (one is a scalar multiple of the other).

Definition (Angle Between Vectors)

The *angle* $\theta \in [0, \pi]$ between nonzero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is defined by

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|}.$$

The vectors are *orthogonal* (written $\mathbf{v} \perp \mathbf{w}$) if $\theta = \pi/2$, equivalently if $\mathbf{v} \cdot \mathbf{w} = 0$.

Remark (Dot product and normals in NURBS). The unit normal to a parametric surface $S(u, v)$ is computed as $N = S_u \times S_v / \|S_u \times S_v\|$, which requires both the cross product and normalisation via the norm. The dot product also enters the convex hull argument: showing that a B-spline curve lies on a specific side of a hyperplane reduces to checking the sign of dot products of control points with the hyperplane's normal.

4.4 Introduction to Matrices

Matrices and Matrix–Vector Multiplication

Definition (Matrix)

An $m \times n$ *matrix* A is a rectangular array of real numbers with m rows and n columns:

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix}.$$

The entry in row i, column j is denoted A_{ij} or $[A]_{ij}$. We write $A \in \mathbb{R}^{m \times n}$.

Definition (Matrix–Vector and Matrix–Matrix Multiplication)

If $A \in \mathbb{R}^{m\times n}$ and $\mathbf{x} \in \mathbb{R}^n$, then $A\mathbf{x} \in \mathbb{R}^m$ is defined by

$$(A\mathbf{x})_i = \sum_{j=1}^{n} A_{ij}x_j, \qquad i = 1, \dots, m.$$

Equivalently, $A\mathbf{x}$ is the linear combination of the columns of A with coefficients from $\mathbf{x}$:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, their product $AB \in \mathbb{R}^{m\times p}$ has entries

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik}B_{kj}.$$

Remark (Matrix multiplication is not commutative). In general $AB \neq BA$, even when both products are defined. Order matters: applying transformation B first, then A, is written AB.

The Transpose

Definition (Transpose)

The *transpose* of $A \in \mathbb{R}^{m\times n}$ is the matrix $A^T \in \mathbb{R}^{n\times m}$ defined by $(A^T)_{ij} = A_{ji}$. Rows of A become columns of A^T. Key properties:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$
- $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T\mathbf{w}$ (dot product as matrix multiplication)

Matrix Form of a Curve

Remark (NURBS connection: matrix form of curves and surfaces). The power basis curve $C(u) = \sum_{i=0}^{n} \mathbf{a}_i u^i$ can be written as

$$C(u) = [\mathbf{a}_0 \;\; \mathbf{a}_1 \;\; \cdots \;\; \mathbf{a}_n] \begin{pmatrix} 1 \\ u \\ \vdots \\ u^n \end{pmatrix} = A\,\mathbf{u}(u),$$

where A is the matrix of coefficient vectors and $\mathbf{u}(u)$ is the monomial vector. A Bézier curve has the equivalent matrix form $C(u) = [u^i]^T M[P_i]$, where M is the $(n+1) \times (n+1)$ change-of-basis matrix converting Bernstein basis to power basis. A tensor product surface takes the form $S(u, v) = [f_i(u)]^T [b_{i,j}][g_j(v)]$, a matrix sandwiched between two basis-function vectors. All of these representations rely on matrix–vector multiplication as defined above.

4.5 Solving Linear Equations

Matrix Operations — Quick Reference

Associativity	$A(BC) = (AB)C$.
Distributivity	$A(B+C) = AB + AC$; $(A+B)C = AC + BC$.
NOT commutative	$AB \neq BA$ in general.
Identity	$AI = IA = A$.
Transpose product	$(AB)^T = B^T A^T$.
Diagonal matrices	$D_1 D_2$ is diagonal; DA scales rows; AD scales columns.

Inverse Matrices — Quick Reference

Inverse	A^{-1}: the unique matrix with $AA^{-1} = A^{-1}A = I$.
Existence	A^{-1} exists iff A is square and $\det(A) \neq 0$ (nonsingular).
2×2 **formula**	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
Product rule	$(AB)^{-1} = B^{-1}A^{-1}$ (order reverses).
Transpose rule	$(A^T)^{-1} = (A^{-1})^T$.
Change of basis	If M converts basis B_1 to B_2, then M^{-1} converts B_2 to B_1.

4.6 Rules for Matrix Operations

Algebraic Rules

Theorem (Rules for Matrix Multiplication)

When the dimensions are compatible:

1. $A(BC) = (AB)C$ (associativity)
2. $A(B + C) = AB + AC$ (left distributivity)
3. $(A + B)C = AC + BC$ (right distributivity)
4. $c(AB) = (cA)B = A(cB)$ for any scalar c
5. $(AB)^T = B^T A^T$ (transpose reverses order)

The following do **not** hold in general:

- $AB \neq BA$ (not commutative)
- $AB = AC \not\Rightarrow B = C$ (no cancellation in general)
- $AB = 0 \not\Rightarrow A = 0$ or $B = 0$

Diagonal Matrices

Definition (Diagonal Matrix)

A *diagonal matrix* $D \in \mathbb{R}^{n\times n}$ has $D_{ij} = 0$ for all $i \neq j$. Written $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$.
Key computational facts:

- DA multiplies each *row* i of A by d_i.
- AD multiplies each *column* j of A by d_j.
- $D_1D_2 = \operatorname{diag}(d_1e_1, \ldots, d_ne_n)$ where e_j are the diagonal entries of D_2.

Block Matrices

Remark (Block matrix multiplication). Matrices can be partitioned into blocks, and multiplied block-by-block as if the blocks were scalar entries, provided the block sizes are compatible. If $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ and $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$, then

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix},$$

provided inner block dimensions match. The standard matrix of a linear transformation $[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)]$ is a block matrix with column vectors as blocks, and $[T]\mathbf{v} = v_1T(\mathbf{e}_1) + \cdots + v_nT(\mathbf{e}_n)$ is block matrix–vector multiplication.

4.7 Inverse Matrices

Definition and Properties

Definition (Invertible Matrix)

A square matrix $A \in \mathbb{R}^{n\times n}$ is *invertible* (or *nonsingular*) if there exists a matrix $A^{-1} \in \mathbb{R}^{n\times n}$ such that

$$AA^{-1} = A^{-1}A = I_n.$$

The matrix A^{-1} is the *inverse* of A and is unique. A matrix with no inverse is called *singular*.

Theorem (Properties of Inverses)

When A and B are invertible $n \times n$ matrices:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$ (order reverses)
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(cA)^{-1} = \frac{1}{c}A^{-1}$ for $c \neq 0$

The 2×2 Case

Formula (2×2 **Inverse**)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad ad - bc \neq 0.$$

The quantity $ad - bc$ is the determinant. The matrix is singular when $ad = bc$.

Change of Basis via Inverse

Remark (NURBS connection: converting between Bernstein and power basis). In Piegl & Tiller Exercise 1.14, the Bernstein basis functions of degree n can be written as $[B_{0,n}(u), \ldots, B_{n,n}(u)] = [1, u, \ldots, u^n]M$, where M is an $(n+1) \times (n+1)$ matrix. This means a Bézier curve has the matrix form

$$C(u) = [u^i]^T M [P_i],$$

where $[u^i]^T$ is the monomial row vector, $[P_i]$ is the column of control points, and M encodes the change from power basis to Bernstein basis.

The inverse M^{-1} converts the other direction: if you have a curve in power basis form with coefficients $[\mathbf{a}_i]$, then the equivalent Bézier control points are $[P_i] = M^{-1}[\mathbf{a}_i]$.

This is a direct application of the matrix inverse: M converts one basis to another, and M^{-1} inverts the conversion.

4.8 Vector Spaces

Basis and Dimension — Quick Reference

Linearly independent	No vector is a linear combination of the others; only $\mathbf{0} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$ has $c_i = 0$ for all i.
Span	$\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$: all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_k$.
Basis	A linearly independent set that spans the space.
Dimension	Number of vectors in any basis; all bases have the same size.
Change of basis	Transition matrix P: columns are the new basis vectors in terms of the old basis.

4.9 Independence, Basis, and Dimension

Linear Independence

Definition (Linear Independence)

Vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ are *linearly independent* if the only solution to

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

is $c_1 = c_2 = \cdots = c_k = 0$. Otherwise they are *linearly dependent*: at least one vector is a linear combination of the others.

Basis and Dimension

Definition (Basis)

A *basis* for a subspace $V \subseteq \mathbb{R}^n$ is a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ that is:

1. linearly independent, and
2. spans V (every vector in V is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$).

Theorem (Dimension is Well-Defined)

Any two bases for the same subspace V have the same number of vectors. This common number is called the *dimension* of V, written $\dim V$.

Remark. The standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$, so $\dim \mathbb{R}^n = n$. The key insight: every set of n linearly independent vectors in $\mathbb{R}^n$ is automatically a basis.

Change of Basis

Definition (Transition Matrix)

Let $\mathcal{B}_1 = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ and $\mathcal{B}_2 = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be two ordered bases for $\mathbb{R}^n$. The *transition matrix* P from $\mathcal{B}_1$ to $\mathcal{B}_2$ has as its i-th column the coordinates of $\mathbf{u}_i$ expressed in the basis $\mathcal{B}_2$. If $[\mathbf{x}]_{\mathcal{B}_1}$ denotes the coordinate vector of $\mathbf{x}$ in basis $\mathcal{B}_1$, then

$$[\mathbf{x}]_{\mathcal{B}_2} = P\,[\mathbf{x}]_{\mathcal{B}_1}.$$

The inverse P^{-1} converts in the opposite direction.

Remark (NURBS connection: Bernstein basis as a basis for polynomials)*.* The Bernstein polynomials $\{B_{0,n}(u), B_{1,n}(u), \ldots, B_{n,n}(u)\}$ form a basis for the vector space of polynomials of degree $\leq n$. So does the monomial (power) basis $\{1, u, u^2, \ldots, u^n\}$. The matrix M from Piegl & Tiller Exercise 1.14 is precisely the transition matrix converting coordinates in the monomial basis to coordinates in the Bernstein basis. M^{-1} converts the other direction. This is why the Bézier curve $C(u) = [u^i]^T M [P_i]$ can be read as: "express the Bernstein basis functions in the monomial basis via M, then multiply by the control points."

4.10 Orthogonality

4.11 Determinants

Determinants and Cross Product — Quick Reference

2×2 **det**	$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$
3×3 **det**	Cofactor expansion along any row or column.
Cross product	$\mathbf{v} \times \mathbf{w}$: vector orthogonal to both $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$, computed via 3×3 determinant mnemonic.
Magnitude	$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \, \|\mathbf{w}\| \sin\theta$; equals area of parallelogram.
Direction	Right-hand rule; $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$.

4.12 Determinants and the Cross Product

The 3×3 Determinant

Definition (3×3 Determinant via Cofactor Expansion)

For $A \in \mathbb{R}^{3\times 3}$, expanding along the first row:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

The signs alternate: $+, -, +$ along the first row.

The Cross Product

Definition (Cross Product)

For $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ in $\mathbb{R}^3$, the *cross product* is

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2w_3 - v_3w_2, \;\; v_3w_1 - v_1w_3, \;\; v_1w_2 - v_2w_1).$$

Theorem (Properties of the Cross Product)

For $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$ and $c \in \mathbb{R}$:

1. $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$ (anticommutativity)
2. $\mathbf{v} \times (\mathbf{w} + \mathbf{x}) = \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x}$ (distributivity)
3. $\mathbf{v} \times \mathbf{v} = \mathbf{0}$
4. $(c\mathbf{v}) \times \mathbf{w} = c(\mathbf{v} \times \mathbf{w})$
5. $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ and $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ (orthogonality)
6. $\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \, \|\mathbf{w}\| \sin\theta$ (magnitude = parallelogram area)

Remark (NURBS connection: surface normal vectors)*.* The unit normal to a parametric surface at a point (u, v) is

$$N(u, v) = \frac{S_u \times S_v}{\|S_u \times S_v\|},$$

where S_u and S_v are the partial derivative vectors of the surface with respect to the parameters. The cross product gives the direction orthogonal to both tangent directions; dividing by the norm produces the unit normal. This appears in Piegl & Tiller §1.1 and is used throughout whenever normal vectors are needed (shading, offset surfaces, etc.).

Remark (Mnemonic for the cross product)*.* Write the entries of $\mathbf{v}$ and $\mathbf{w}$ each twice in a 2×6 array. Ignore the first and last columns. Draw three "X" shapes between columns 2–3, 3–4, and 4–5. Each "X" contributes one entry of $\mathbf{v} \times \mathbf{w}$: add along forward diagonals, subtract along backward diagonals.

4.13 Eigenvalues

4.14 Singular Value Decomposition

4.15 Linear Transformations

Standard Matrix — Quick Reference

Standard matrix	$[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \in \mathbb{R}^{m\times n}$.
How to build it	Evaluate T on each standard basis vector; place results as columns.
Uniqueness	$[T]$ is the *unique* matrix with $T(\mathbf{v}) = [T]\mathbf{v}$ for all $\mathbf{v}$.
Composition	$[S \circ T] = [S][T]$.
Change of basis	Same T, different bases $\Rightarrow$ different matrix; related by $B = P^{-1}AP$.

Linear Transformations — Quick Reference

Definition	$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w}$ and scalars c, d.
Origin fixed	Always $T(\mathbf{0}) = \mathbf{0}$; translation is *not* linear.
Standard matrix	$[T] = [T(\mathbf{e}_1) \mid \cdots \mid T(\mathbf{e}_n)]$; then $T(\mathbf{v}) = [T]\mathbf{v}$.
Composition	$S \circ T$ has matrix $[S][T]$.
Affine map	$\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$: linear part A plus translation $\mathbf{v}$.
Invariance	B-spline curves: apply affine map to control points only.

4.16 The Idea of a Linear Transformation

Definition

Definition (Linear Transformation)

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a *linear transformation* if

$$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$$

for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and all $c, d \in \mathbb{R}$.
Equivalently, T must satisfy both:

1. $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ (preserves addition), and
2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (preserves scalar multiplication).

Remark (The origin is always fixed). Every linear transformation satisfies $T(\mathbf{0}) = \mathbf{0}$. Therefore, *translation* (shifting by a nonzero vector) is not linear. The four geometric operations that *are* linear: rotation about the origin, scaling, reflection, and projection.

Geometric Catalog in $\mathbb{R}^2$

Standard Planar Transformations

Scaling	$\begin{pmatrix} c_1 & 0 \\ 0 & c_2 \end{pmatrix}$: scales x by c_1, y by c_2.
Rotation by θ	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$: rotates counter-clockwise.
Reflection ($y = x$)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: swaps x and y coordinates.
Projection (x-axis)	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$: drops the y component.

Remark (Composition is matrix multiplication). Applying T_1 then T_2 has matrix $[T_2][T_1]$. A sequence of geometric operations on a model (scale, then rotate, then translate) corresponds to multiplying their matrices together in that order.

Affine Transformations

Definition (Affine Transformation)

An *affine transformation* $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ has the form

$$\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v},$$

where $A \in \mathbb{R}^{n\times n}$ is the linear part and $\mathbf{v} \in \mathbb{R}^n$ is a translation vector. An affine transformation is a linear transformation when $\mathbf{v} = \mathbf{0}$.

Remark (Affine invariance of B-spline curves — NURBS Property P3.4). An affine transformation $\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$ applied to a B-spline curve $C(u) = \sum_{i=0}^{n} N_{i,p}(u)P_i$ gives

$$\Phi(C(u)) = \sum_{i=0}^{n} N_{i,p}(u)\,\Phi(P_i).$$

The transformation can be applied to the curve by applying it to each control point individually. This works because the basis functions sum to 1 (partition of unity):

$$\Phi(C(u)) = A\Big(\sum N_{i,p}P_i\Big) + \mathbf{v} = \sum N_{i,p}(AP_i) + \underbrace{\Big(\sum N_{i,p}\Big)}_{=1}\mathbf{v} = \sum N_{i,p}(AP_i + \mathbf{v}).$$

This means rotating, translating, or scaling a B-spline curve costs only $n + 1$ affine operations on control points, regardless of how many points are evaluated on the curve.

4.17 The Matrix of a Linear Transformation

The Standard Matrix

Theorem (Standard Matrix of a Linear Transformation)

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there exists a unique matrix $[T] \in \mathbb{R}^{m\times n}$ such that

$$T(\mathbf{v}) = [T]\mathbf{v} \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

The *standard matrix* $[T]$ is constructed by placing the images of the standard basis vectors as columns:

$$[T] = \big[\, T(\mathbf{e}_1) \;\mid\; T(\mathbf{e}_2) \;\mid\; \cdots \;\mid\; T(\mathbf{e}_n) \,\big].$$

Remark (Why this works). Since every $\mathbf{v} = v_1\mathbf{e}_1 + \cdots + v_n\mathbf{e}_n$, linearity gives

$$T(\mathbf{v}) = v_1T(\mathbf{e}_1) + \cdots + v_nT(\mathbf{e}_n) = [T]\mathbf{v}.$$

Knowing T on the basis vectors determines T on the entire space.

Matrix Under Change of Basis

Definition (Matrix Relative to a Basis)

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation and let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be an ordered basis. The *matrix of T relative to $\mathcal{B}$*, denoted $[T]_\mathcal{B}$, has as its j-th column the coordinates of $T(\mathbf{b}_j)$ expressed in the basis $\mathcal{B}$.
If P is the transition matrix whose columns are $\mathbf{b}_1, \dots, \mathbf{b}_n$ expressed in the standard basis, then

$$[T]_\mathcal{B} = P^{-1}[T]P.$$

Two matrices related by $B = P^{-1}AP$ are called *similar*.

Remark (NURBS connection: Bézier curve in matrix form). The Bézier curve has the matrix representation

$$C(u) = [u^i]^T M[P_i],$$

where:

- $[u^i]^T = (1, u, u^2, \dots, u^n)$ is the monomial basis vector,
- $[P_i]$ is the column vector of control points,
- M is the $(n + 1) \times (n + 1)$ matrix that encodes the Bernstein polynomials in the monomial basis.

The matrix M is the standard matrix of the linear transformation that converts from Bernstein basis coordinates to monomial basis coordinates. Setting $[\mathbf{a}_i] = M[P_i]$ gives the power basis representation of the same curve (so the Bézier coefficients $[P_i]$ are the coordinates of the curve in the Bernstein basis, and $[\mathbf{a}_i]$ are the coordinates in the monomial basis). Converting between the two is exactly the change-of-basis operation $P^{-1}AP$ applied to the coefficient vectors.

4.18 Complex Vectors and Matrices

4.19 Applications

4.20 Numerical Linear Algebra

4.21 Probability and Statistics

Chapter 5

Computational Probability

computational-probability

Computational Linear Algebra $\rightarrow$ **Computational Probability** $\rightarrow$
Ordinary Differential Equations

5.1 Computational Probability

Remark (Planned content). Active mathematical content has not yet been added.

Chapter 6

Ordinary Differential Equations

ordinary-differential-equations

Computational Probability → **Ordinary Differential Equations** → Partial Differential Equations

Where This Chapter Lives

Calculus → **Ordinary Differential Equations** → Dynamical Systems / Numerical Methods / PDE

Why this chapter belongs here. Ordinary differential equations are where derivatives stop being only local rates and start governing whole families of evolving states. Once differentiation is in hand, equations such as

$$y'(t) = f(t, y(t))$$

become the natural language for motion, growth, decay, forcing, and feedback.

What this chapter will gather. The eventual ODE chapter is the right home for:

- first-order equations and separable models;
- linear equations and integrating factors;
- existence and uniqueness ideas;
- systems of ODEs and phase portraits;
- stability, equilibrium, and qualitative behavior;
- numerical methods such as Euler and Runge–Kutta schemes.

Why this is currently a breadcrumb. The surrounding curriculum is still building the supporting architecture: calculus, linear algebra, and numerical analysis. ODEs sit at the meeting point of all three, so this file marks the destination clearly even before the chapter is fully written.

6.1 Models

Definition

Definition 6.1.1 (Differential Equation)**.** A **differential equation** is an equation whose unknown is a function and whose terms involve one or more derivatives of that function.
For an unknown function $y = y(t)$, a differential equation may be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0.$$

Here t is the independent variable, $y(t)$ is the unknown function, and

$$y', y'', \dots, y^{(n)}$$

are derivatives of y.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* At the computational level, a differential equation describes how a quantity changes. Instead of giving the function $y(t)$ directly, it gives a relation involving the rate of change of $y(t)$, and possibly higher-order rates of change.

Example (A First Differential Equation)*.* The equation

$$y' = 3y$$

is a differential equation because it relates the unknown function y to its derivative y'.

A solution is

$$y(t) = Ce^{3t},$$

where C is a constant. Indeed,

$$y'(t) = 3Ce^{3t} = 3y(t).$$

Definition

Definition 6.1.2 (Order of a Differential Equation)**.** The **order** of a differential equation is the highest derivative of the unknown function that appears in the equation.
For example, if an equation involving an unknown function $y = y(t)$ can be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0$$

and $y^{(n)}$ is the highest derivative that appears, then the differential equation has **order** n.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Example (Orders of Differential Equations)*.* The equation

$$y' = 3y$$

is first order, since the highest derivative appearing is y'.

The equation

$$y'' + 4y' + 5y = 0$$

is second order, since the highest derivative appearing is y''.

The equation

$$y^{(4)} - ty'' + y = 0$$

is fourth order, since the highest derivative appearing is $y^{(4)}$.

Definition

Definition 6.1.3 (Solution of a Differential Equation)**.** A **solution** of a differential equation is a function that satisfies the equation on a specified interval.
More precisely, if a differential equation is written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0,$$

then a solution on an interval I is a function $y : I \to \mathbb{R}$ such that y has all derivatives required by the equation and

$$F(t, y(t), y'(t), y''(t), \ldots, y^{(n)}(t)) = 0$$

for every $t \in I$.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* A solution is not a number; it is a function. The function solves the differential equation when substituting the function and its derivatives into the equation makes the equation true for every point in the interval under consideration.

Example (Checking a Solution)*.* The function

$$y(t) = Ce^{3t}$$

is a solution of

$$y' = 3y$$

on $\mathbb{R}$, because

$$y'(t) = 3Ce^{3t} = 3y(t)$$

for every $t \in \mathbb{R}$.

Definition

Definition 6.1.4 (Autonomous Differential Equation)**.** An **autonomous differential equation** is a differential equation in which the independent variable does not appear explicitly.
For a first-order ordinary differential equation, an autonomous equation has the form

$$y' = f(y),$$

where the rate of change of y depends only on the current value of y, not directly on time t.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* An autonomous differential equation describes a system whose rule of evolution does not change with time. If the system is in the same state y at two different times, then the differential equation assigns the same rate of change at both times.

Example (Autonomous Equations)*.* The equation

$$y' = 3y$$

is autonomous, since the right-hand side depends only on y.

The equation

$$y' = t + y$$

is not autonomous, since the independent variable t appears explicitly.

Definition

Definition 6.1.5 (Non-Autonomous Differential Equation)**.** A **non-autonomous differential equation** is a differential equation in which the independent variable appears explicitly.

For a first-order ordinary differential equation, a non-autonomous equation has the form

$$y' = f(t, y),$$

where the rate of change of y may depend both on the current value of y and on the independent variable t.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* A non-autonomous differential equation describes a system whose rule of evolution may change over time. Even if the system has the same state y at two different times, the differential equation may assign different rates of change because the value of t may be different.

Example (Non-Autonomous Equations)*.* The equation

$$y' = t + y$$

is non-autonomous, since the independent variable t appears explicitly.

The equation

$$y' = 3y$$

is autonomous, not non-autonomous, since the right-hand side depends only on y.

Definition

Definition 6.1.6 (Linear Differential Equation)**.** A differential equation is called **linear** if the unknown function and its derivatives appear only to the first power and are not multiplied together.
An nth-order linear ordinary differential equation for an unknown function $y = y(t)$ has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

where the coefficient functions

$$a_0(t), a_1(t), \ldots, a_n(t)$$

and the forcing term $g(t)$ are given functions of the independent variable t.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* Linearity means that the equation is linear in the unknown function y and its derivatives. The coefficients may depend on t, but they may not depend on y or its derivatives. Thus terms such as y^2, yy', $\sin(y)$, and $(y')^2$ make an equation nonlinear.

Example (Linear and Nonlinear Equations)*.* The equation

$$y'' + 3ty' + 2y = \sin t$$

is linear.

The equation

$$y'' + y^2 = 0$$

is nonlinear because it contains the term y^2.

The equation

$$y' + yy' = t$$

is nonlinear because it contains the product yy'.

Ordinary differential equations are often used to model physical phenomena because many natural systems are governed by relationships between a quantity and its rate of change. In such models, the unknown function represents the state of the system, while the differential equation describes how that state evolves over time. For example, population models describe how a population changes, cooling models describe how temperature changes, motion models describe how position and velocity change, and circuit models describe how current or voltage changes. In each case, the differential equation does not usually give the quantity directly; instead, it gives a rule for how the quantity changes.

Definition

Definition 6.1.7 (Malthusian Population Growth Model)**.** The **Malthusian population growth model** assumes that the rate of change of a population is proportional to the current population.
If $P(t)$ denotes the population at time t, then the model is

$$P'(t) = kP(t),$$

where k is a constant called the **growth rate** or **proportionality constant**.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* The model says that population growth depends only on the current population size. If $k > 0$, the population grows exponentially. If $k < 0$, the population decays exponentially. If $k = 0$, the population remains constant.

Example (General Solution)*.* The general solution of

$$P'(t) = kP(t)$$

is

$$P(t) = Ce^{kt},$$

where C is a constant. If the initial population is $P(0) = P_0$, then

$$P(t) = P_0e^{kt}.$$

Definition

Definition 6.1.8 (Decay)**.** A quantity is said to undergo **decay** if it decreases over time.
In a differential equation model, if $Q(t)$ denotes the quantity at time t, then decay is often modeled by an equation of the form

$$Q'(t) = -kQ(t),$$

where $k > 0$ is a constant.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* The equation

$$Q'(t) = -kQ(t)$$

says that the rate of change of the quantity is proportional to the current amount, but with a negative sign. Thus larger amounts decay faster in absolute size, while smaller amounts decay more slowly.

Example (Exponential Decay)*.* The general solution of

$$Q'(t) = -kQ(t)$$

is

$$Q(t) = Ce^{-kt}.$$

If $Q(0) = Q_0$, then

$$Q(t) = Q_0 e^{-kt}.$$

For $k > 0$, this function decreases toward 0 as t increases.

Definition

Definition 6.1.9 (Newton's Law of Cooling)**. Newton's Law of Cooling** states that the rate at which an object's temperature changes is proportional to the difference between the object's temperature and the surrounding ambient temperature.
If $T(t)$ denotes the temperature of the object at time t, and T_a denotes the ambient temperature, then the model is

$$T'(t) = -k\big(T(t) - T_a\big),$$

where $k > 0$ is a constant called the **cooling constant**.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* The quantity

$$T(t) - T_a$$

measures how far the object's temperature is from the ambient temperature. Newton's Law of Cooling says that the larger this temperature difference is, the faster the object cools or warms toward the ambient temperature.

Example (General Solution)*.* The general solution of

$$T'(t) = -k\big(T(t) - T_a\big)$$

is

$$T(t) = T_a + Ce^{-kt},$$

where C is a constant. If $T(0) = T_0$, then

$$T(t) = T_a + (T_0 - T_a)e^{-kt}.$$

Definition

Definition 6.1.10 (General Solution)**.** A **general solution** of a differential equation is a family of solutions containing one or more arbitrary constants.
For an nth-order ordinary differential equation, the general solution typically contains n arbitrary constants:

$$y(t) = \Phi(t, C_1, C_2, \dots, C_n).$$

Each choice of the constants $C_1, C_2, \dots, C_n$ gives a particular solution of the differential equation.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). The general solution represents the full family of solutions produced by solving the differential equation. Initial conditions are then used to choose specific values of the arbitrary constants, producing a particular solution.

Example (General Solution). The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t},$$

where C is an arbitrary constant. If an initial condition such as

$$y(0) = 5$$

is imposed, then $C = 5$, and the particular solution is

$$y(t) = 5e^{3t}.$$

Definition

Definition 6.1.11 (Family of Solutions)**.** A **family of solutions** of a differential equation is a collection of solution functions described by one or more parameters. For example, a family of solutions may be written in the form

$$y(t) = \Phi(t, C),$$

or more generally,

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n),$$

where $C, C_1, \ldots, C_n$ are parameters. Each choice of the parameters gives one particular solution.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). A family of solutions records many solutions at once. The parameters act as labels that distinguish one solution curve from another. When initial conditions are supplied, they usually determine the parameter values and select a single member of the family.

Example (A Family of Solutions). The differential equation

$$y' = 3y$$

has the family of solutions

$$y(t) = Ce^{3t},$$

where $C \in \mathbb{R}$. Each value of C gives a different solution.

Definition

Definition 6.1.12 (Initial Condition)**.** An **initial condition** specifies the value of the unknown function at a particular value of the independent variable.
For a first-order differential equation involving an unknown function $y = y(t)$, an initial condition has the form

$$y(t_0) = y_0,$$

where t_0 is the initial time and y_0 is the initial value.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). An initial condition selects one solution from a family of solutions. If a differential equation has general solution

$$y(t) = \Phi(t, C),$$

then the condition

$$y(t_0) = y_0$$

is used to determine the constant C.

Example (Initial Condition). The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t}.$$

If the initial condition is

$$y(0) = 5,$$

then

$$5 = Ce^0 = C.$$

Thus the particular solution satisfying the initial condition is

$$y(t) = 5e^{3t}.$$

Definition

Definition 6.1.13 (Initial Value Problem)**.** An **initial value problem**, or **IVP**, is a differential equation together with one or more initial conditions.
For a first-order differential equation, an initial value problem has the form

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

The differential equation describes how the function changes, while the initial condition specifies the value of the solution at the initial time t_0.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). An initial value problem asks for the particular solution of a differential equation that passes through a specified point. Geometrically, the condition

$$y(t_0) = y_0$$

means that the solution curve must pass through the point

$$(t_0, y_0).$$

Example (Initial Value Problem). Consider the initial value problem

$$y' = 3y, \qquad y(0) = 5.$$

The differential equation has general solution

$$y(t) = Ce^{3t}.$$

Using the initial condition,

$$5 = y(0) = Ce^{0} = C.$$

Therefore the solution of the initial value problem is

$$y(t) = 5e^{3t}.$$

Definition

Definition 6.1.14 (Existence). An initial value problem is said to have **existence** if there is at least one function that satisfies both the differential equation and the initial condition.
For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

existence means that there is a function $y(t)$, defined on some interval containing t_0, such that

$$y'(t) = f(t, y(t))$$

and

$$y(t_0) = y_0.$$

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Definition

Definition 6.1.15 (Uniqueness). An initial value problem is said to have **uniqueness** if it has at most one solution on the interval under consideration.
For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

uniqueness means that if $y_1(t)$ and $y_2(t)$ are both solutions on the same interval I containing t_0, then

$$y_1(t) = y_2(t)$$

for every $t \in I$.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Definition

Definition 6.1.16 (Interval of Existence)**.** An **interval of existence** for a solution of an initial value problem is an interval on which the solution is defined and satisfies the differential equation.
For an initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

an interval I is an interval of existence if $t_0 \in I$ and there is a solution $y : I \to \mathbb{R}$ such that

$$y'(t) = f(t, y(t))$$

for every $t \in I$, and

$$y(t_0) = y_0.$$

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* Existence asks whether at least one solution passes through the initial point. Uniqueness asks whether only one solution passes through that point. The interval of existence records how long the solution is valid.

Example (Existence and Uniqueness)*.* The initial value problem

$$y' = 3y, \qquad y(0) = 5$$

has the solution

$$y(t) = 5e^{3t}.$$

This solution exists and is unique on $\mathbb{R}$.

The interval of existence is

$$(-\infty, \infty).$$

Chapter 7

Partial Differential Equations

partial-differential-equations

Ordinary Differential Equations $\rightarrow$ **Partial Differential Equations** $\rightarrow$ Fourier Analysis

7.1 Partial Differential Equations

Remark (Planned content). Active mathematical content has not yet been added.

Chapter 8

Fourier Analysis

fourier-analysis

Partial Differential Equations → **Fourier Analysis** → Numerical Analysis

Bibliography